

Connor Aird

Research Software Engineer at UCL
London, United Kingdom

Skills

`</>` C++, Fortran, Python
`>_` Docker, Git, Qt, Quarto
 MPI, OpenMP

Education

09/2016 - 06/2020

MPhys Physics

University of York

1st Class Honours (with distinction)

Awards

2026

CAKE Fellowship

Awarded funding for developing and delivering Fortran Unit Testing workshops across the UK

2026

Qualified Carpentries Instructor

Experience teaching UNIX shell, Python, Git and Fortran

Contact

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 connoraird.github.io
 connoraird
 connoraird

Experience

Research Software Engineer

08/2024 - Present

Centre for Advanced Research Computing
University College London

Beyond my Assistant RSE responsibilities, I now manage projects including staffing and PI collaboration. I develop and deliver workshop training materials, and regularly present at conferences such as RSECon and FOSDEM.

Assitant Research Software Engineer

06/2023 - 08/2024

Centre for Advanced Research Computing
University College London

Collaborated with researchers across UCL to design, develop, improve and maintain research codes, advised on software best practices, and supported Carpentries teaching and marking.

Consultant

08/2020 - 06/2023

Accenture
Leeds, UK

Built well-tested web applications, wrote AWS infrastructure-as-code using Terraform and maintained CI/CD pipelines with Jenkins and GitHub Actions.

Projects

GLASS, Porting to the python array API

09/2025 - Present

RSE, UCL

Ported existing functions utilising NumPy, within this Python codebase, to be array api compatible and created a benchmark suite using pytest benchmark.

k-Plan Treatment Planning Module

07/2023 - Present

RSE, UCL

Schedule staffing on the project, Prioritise work, Implement feature requests, bug fixes and general maintenance and advise on implementation and design.

MUSIC, DiRAC GPU feasibility study

04/2025 - 06/2025

RSE, UCL

Aimed to determine the potential gains and likely cost of GPU offloading the MUSIC codebase. Produced scripts for building and generating benchmark data for MUSIC. Also profiled MUSIC using the NVIDIA NSIGHT system.

CONQUEST, eCSE08 Improving Multi-threaded Scaling.

01/2024 - 06/2024

RSE, UCL

Refactored a kernel within this Fortran code to maximise serial performance, implemented OpenMP and BLAS parallelisation and carried out performance analysis using benchmarks and profilers to tune this parallel implementation.

Professional Development

02/2026

Fortran Course

Developed and instructed my own course on unit testing Fortran.

10/2025 - Present

TechSocials

Organiser of the TechSocials series.

09/2024 - Present

Fortran collaboration

Fortran community collaboration (grant applications, hackathons, etc)